Dynamic Market Making Under Stochastic Liquidity and Inventory Risk

An Event-Driven Evaluation with Depth-Conditioned Adverse Selection

Abstract

We study whether adding self-exciting order flow and stochastic liquidity to a finite-horizon market-making controller improves established inventory-aware benchmarks, and whether the adverse-selection input used by the controller is aligned with the fills it is intended to price. The empirical analysis uses Binance BTCUSDT and ETHUSDT perpetual-futures aggregate trades and top-of-book updates. First, a separately frozen measurement study estimates cumulative reach-conditioned maker markouts: an aggressive event capable of reaching a deep quote also belongs to the exposure set of every shallower quote. The 10-second adverse-markout difference between quotes at 1.5 and 0.5 ticks is positive on all 47 original BTCUSDT days, averaging 31.38 ticks in validation (95% confidence interval [30.21, 32.56]) and 37.76 ticks in the final holdout ([35.09, 40.44]). A prospectively frozen 20-day extension passes separately for BTCUSDT (42.96 ticks; [39.67, 46.26]) and ETHUSDT (22.47 ticks; [20.57, 24.37]). Second, we compare eight quoting policies under common realized event reach and common queue-capture draws. A full Hamilton–Jacobi–Bellman (HJB) controller, conditioned on causal buy and sell Hawkes intensities and a three-state liquidity regime, improves stationary Guéant–Lehalle–Fernandez-Tapia (GLFT) by 0.105 USDT per hour (familywise interval [0.017, 0.193]) but underperforms classical Avellaneda–Stoikov (AS) by 0.496 USDT per hour ([−0.742, −0.251]). All absolute market-making PnLs (Profit and Loss) are negative. The HJB selects its maximum tested depth on at least one side in 96.7% of quote intervals, while its adverse-selection input is scalar in depth and the fill-decay level is identified from event reach rather than queue fills. The combined evidence does not establish a profitable strategy or a corrected controller. It shows that richer state conditioning alone does not provide robust benchmark improvement when execution opportunity and fill-conditional toxicity are not estimated at the same depth and event resolution.

Keywords: *market making; Avellaneda–Stoikov; Hawkes process; stochastic liquidity; Hamilton–Jacobi–Bellman equation; adverse selection; limit order book; preregistered inference*

JEL Classification: *G12; G14; C51; C58*

1. Introduction

A market maker repeatedly chooses how far to quote from a reference price while balancing spread capture, uncertain execution, adverse selection, and inventory risk. Avellaneda and Stoikov (2008) turned this trade-off into a clean stochastic-control problem. Guéant, Lehalle, and Fernandez-Tapia (2013) later built a simpler stationary version of this, and it is now a standard benchmark for inventory-aware quoting. These models are deliberately simplified. Order flow is usually modeled as Poisson arrivals, with intensity that decays as quote depth increases. The cost of trading against an informed order is either left implicit, or squeezed into a small penalty term.

Two extensions are considered in this study. First, high-frequency order flow tends to cluster. A Hawkes process can capture this and give us a causal forecast of near-term buy and sell activity. Second, liquidity itself is not constant. It shifts with volatility, activity, spread, and queue depth. A Markov regime lets the controller adjust when these conditions change. We can combine both extensions into a finite-state HJB model, along with inventory risk, a no-quote option, and an explicit cost for liquidating at the end.

However, a greater state complexity does not automatically mean a better-identified problem. The key issue is that a passive fill is not the same thing as a random aggressive trade. It is an event that is capable of reaching the maker's chosen price, and only if the quote is still there, and near enough to the front of the queue. If events that reach deeper prices come with larger adverse moves, then something breaks. A model where the fill rate decays with depth, but the toxicity cost stays flat, presents the optimizer with an inconsistent trade-off. The model may imply that quoting further away reduces execution probability, without believing it changes the loss on the rare fill that does happen. This introduces a potential model-specification inconsistency.

This paper connects two separate questions. First: how should we actually measure adverse maker markouts for a hypothetical quote at depth d? Second: does a state-aware HJB model, conditioned on Hawkes order flow and shifting liquidity, actually beat strong AS and GLFT benchmarks, when every policy sees the exact same market events?

These two analyses were not planned together from the start as one experiment. The depth measurement went through its own frozen validation, holdout, and extension stages. The strategy evaluation was frozen separately, before we computed any PnL rankings. But part of its March data window had already been touched by the earlier measurement study. So we describe the strategy design as outcome-blind with respect to the policy rankings we report, not as a completely untouched market holdout. That distinction matters, and we note this explicitly.

The measurement result points in one stable direction. A batch of aggressive trades reaching 1.5 ticks has a much larger maker-signed markout than the full set of events reaching just the 0.5-tick touch. This positive sign holds on 67 straight BTCUSDT days and on all 20 ETHUSDT extension days, even though the size of the effect moves around across time and instruments. The strategy results are reported without imposing a favourable performance criterion. The combined HJB model beats stationary GLFT, but loses to classical AS on both endpoints we pre-specified. Every strategy loses money under our fixed-minute, immediate-replenishment replay.

This result is useful precisely because the benchmark was not built to guarantee a win for our proposed model. We trained forecast parameters before the evaluation period started. Evaluation events only update Hawkes intensity after they happen, never before. Fills are tied to real, realized event reach, not generated from the policy's own internal model. Queue uncertainty comes from common random draws shared across policies. And

we run inference over market days, not over individual events or simulation seeds. The failure mode that shows up is visible and specific. The HJB model is overwhelmingly boundary-seeking. Its depth-decay curve is really an event-reach proxy, not a queue-fill curve. And its adverse-selection input never actually changes with quote depth.

This paper makes three contributions, and we keep them bounded. First, we define a cumulative, reach-conditioned adverse-markout estimand, and we show why exclusive sweep-severity bins fail to represent what a quote is really exposed to. Second, we build a causal, eight-policy benchmark. It separates naive quoting, inventory skew, AS, GLFT, a Hawkes-only HJB, a liquidity-only HJB, and their combined version, all under the same realized events. Third, we connect model inputs back to policy support. A richer controller should not get credit for improvement unless execution opportunity and adverse selection are calibrated at the same quote depth, unless the optimal action actually sits inside the tested range, and unless the policy beats both the stationary and the strong adaptive benchmark.

These claims remain intentionally narrow. Event reach does not reveal actual queue fills. Level-I data do not identify queue rank, cancellation, hidden liquidity, or the lifecycle of a historical resting order. The empirical markout is predictive exposure, not proof of informed trading. The paper does not introduce a post-outcome corrected controller and does not interpret relative replay performance as an investable return.

2. Related Literature

2.1 Inventory-Based Dealer Models

Inventory-based dealer models try to answer one question: how much should a dealer be compensated for holding a risky position? Avellaneda and Stoikov (2008) combine exponential utility, Brownian midprice risk, and Poisson fills that decay with depth. Guéant, Lehalle, and Fernandez-Tapia (2013) give a spectral and asymptotic treatment of the same inventory problem. These two papers supply our main strategy comparators, and they also shape the structure of our own finite HJB model. Cartea, Jaimungal, and Penalva (2015) pull this control-theoretic tradition together into a unified framework covering quoting, execution, and risk design, and their treatment of the finite-horizon value function shapes several choices in this paper, including the explicit terminal liquidation penalty we use in Section 5.3.

More recent work pushes this inventory-control approach toward richer, higher-dimensional settings. Bergault and Guéant (2021) study dealer market making across multiple simultaneous quote requests, using dimensionality-reduction techniques for the resulting value function. Barzykin, Bergault, and Guéant (2023) extend inventory-risk control to multi-currency dealer books. They show that the basic comparative statics of the single-asset AS/GLFT family stay robust even as the state space grows, though the underlying systems get much harder to solve and calibrate. Our own HJB model is comparatively low-dimensional, but it inherits the same caution these papers raise: adding state variables changes what a model can represent, but it does not automatically change what the model can profitably do. We come back to this directly in Section 7.

2.2 Information-Based Microstructure and Adverse Selection

This branch of the literature supplies the other half of the picture: execution cost. Glosten and Milgrom (1985) and Kyle (1985) connect informed order flow, liquidity provision, and price adjustment. Sandås (2001) and Hollifield, Miller, and Sandås (2004) show empirically that execution probability and adverse selection shape limit-order placement together, not separately. Kyle and Obizhaeva (2016) propose a microstructure-invariance

hypothesis, where properly normalized measures of activity, cost, and risk transfer become comparable across assets and time. Our choice to report the depth-conditioned markout separately for BTCUSDT and ETHUSDT, rather than pooling them into one normalized number, follows directly from their warning: naive cross-asset comparisons need an invariance-preserving transformation that Level-I event-reach data simply cannot provide on its own.

Our own contribution here is narrower. We make the exposure set explicit. A quote sitting at depth d is exposed to every aggressive event whose reach is at least d, not just the events whose deepest reach happens to fall in some narrow bin around *d*.

2.3 Self-Exciting Order Flow and Hawkes-Driven Control

Order-flow imbalance tracks closely with short-horizon price changes (Cont, Kukanov, and Stoikov 2014). Hawkes (1971) processes give us a simple way to model event clustering, and Bacry, Mastromatteo, and Muzy (2015) survey how they're used in high-frequency finance. In an earlier companion paper, Bacry, Delattre, Hoffmann, and Muzy (2013) show that a mutually exciting point process can reproduce the microstructure-noise signature we see in real high-frequency prices. That is one reason a self-exciting forecast, rather than a plain stationary Poisson one, is a natural state variable for our controller.

On the control side, Jusselin (2020) works directly on market making under persistent, Hawkes-driven order flow. Using general Hawkes intensities, rather than a simplified exponential decay, that paper proves existence and uniqueness for the associated HJB equation, and proposes a numerical scheme to implement it. Our own finite-state, finite-difference HJB is a simplified variant of that construction. We use bucketed exponential Hawkes intensities as a surface parameter, instead of solving the fully coupled Hawkes-augmented value function, and we flag this simplification directly as a limitation in Section 8.

Alfonsi and Blanc (2016) study a related mixed-market-impact Hawkes price model, and derive optimal execution strategies in closed form, when the price-impact kernel and the order-flow kernel share the same exponential family. That is the same family we use for our fill-decay and intensity kernels. Their results make one thing clear: closed-form tractability in this family depends on the fill-decay curve and the intensity kernel referring to the same event unit and the same depth. That is exactly the alignment condition we show is violated by our deployed controller, in Section 7.2.

Capponi, Figueroa- López, and Yu (2021) model market making with stochastic liquidity demand, forecasting order arrivals and price changes jointly, rather than treating them as separate stylized facts. That is conceptually close to our own three-state liquidity regime, although ours is estimated from training-only quantiles of realized activity, rather than jointly filtered with the price process. Hawkes-driven policies can be richer than stationary Poisson ones, but that richness needs held-out likelihood checks and residual diagnostics to earn any credibility. Order-book covariates may also add information beyond pure self-excitation (Mucciante and Sancetta 2024), which is a reminder that a purely parametric exponential Hawkes intensity is a forecast input, not a complete description of how the market actually works.

2.4 Queue-Reactive and Order-Book Models

Queue-reactive models highlight exactly what Level-I event reach cannot give us. Huang, Lehalle, and Rosenbaum (2015) condition order-book dynamics on queue state directly. Lehalle and Mounjid (2017) study strategic order placement under adverse selection and latency. Law and Viens (2019) show that inconsistent

order-book dynamics can seriously overstate how profitable market making looks. These results motivate our no-quote action, and they're part of why we interpret our replay conservatively.

2.5 Statistical Inference, Multiple Testing, and Preregistration in Strategy Evaluation

A separate strand of the empirical finance literature deals with a different problem: the inference issues that come from testing many candidate strategies, specifications, or endpoints against historical data. Harvey and Liu (2015) frame backtesting as a multiple-hypothesis-testing problem, and show that a reported t-statistic has to be discounted by the number of trials run, whether or not all of those trials get reported, before it can count as real evidence of outperformance. Bailey and López de Prado (2014) propose the deflated Sharpe ratio, which adjusts the significance threshold for a backtested statistic to account for both the number of trials and the non-normality of the underlying returns. That same logic is why we use Bonferroni familywise and Holm-adjusted intervals in Table 5, instead of only reporting the unadjusted paired t-interval for each endpoint. Harvey, Liu, and Zhu (2016) extend this concern to the entire cross-section of proposed return factors, and argue for much higher significance hurdles once you account for every specification that has ever been tried.

We adopt the spirit of this literature procedurally, rather than purely statistically. Both our depth-measurement protocol and our strategy-evaluation protocol were frozen ahead of time. Their estimator, event unit, inference unit, and success rule were all fixed in writing before we opened the corresponding outcome data. This is the preregistration version of controlling the number of trials directly, instead of trying to correct for them after the fact. Section 3.3 and the Data and Reproducibility Statement walk through this chronology in full, and Section 6.2 reports the sensitivity configurations required by our frozen success rule, alongside, not instead of, the primary paired contrasts.

3. Data, Event Units, and Research Sequence

3.1 Source and Samples

The data come from Binance's official public archive for USD-M perpetual futures. We use aggregate-trade messages, bookTicker best-bid and best-ask updates, and one-minute klines for ancillary volatility and bar-level features. Aggressor direction follows is_buyer_maker: false denotes an aggressive buy and true an aggressive sell. Contract ticks are 0.1 USDT for BTCUSDT and 0.01 USDT for ETHUSDT. Download records retain official URLs, byte counts, local SHA-256 values, and exchange checksum verification.

The depth measurement uses the following chronological stages:

Table 1. *Chronological data stages used for the depth-conditioned adverse-markout measurement, by instrument, sample size, and processing volume.*

Stage	Instrument and dates (UTC)	Days	Batches reaching 0.5 ticks
Development	BTCUSDT, 15–21 Jan 2024	7	3,322,373
Validation	BTCUSDT, 22 Jan–10 Feb	20	9,954,350
Final holdout	BTCUSDT, 11 Feb–1 Mar	20	14,143,388
Temporal extension	BTCUSDT, 2–21 Mar	20	20,902,823
Cross-instrument extension	ETHUSDT, 2–21 Mar	20	16,343,707

There are 64,666,641 touch-reachable side/millisecond batches over 87 instrument-days. These counts describe the processing workload and exposure sets; they are not treated as independent observations for inference.

For the strategy study, we use BTCUSDT from 2–30 March. The first ten days, 2–11 March, are the calibration window. The next 19 days, 12–30 March, are the fixed-origin evaluation window. Within one-tick-spread minutes, our preparation layer keeps 14,204 calibration rows and 27,082 evaluation rows, processing 64,336,947 aggregate-trade messages along the way. The measurement counts and the strategy counts differ, because the measurement study follows every eligible event, while the strategy study only constructs fixed minute-start quote opportunities under extra spread and replay conditions.

bookTicker is a Level-I change stream, not a periodic full-depth snapshot. It does not reveal queue position or displayed depth beyond the best quote. The absence of a recent update is not evidence that the book is stale; book-age filters are therefore reported only as declared activity-conditioned sensitivities.

3.2 Identifiable Event Units and Causal Alignment

For the measurement study, our identifiable observation is one aggressor-side/millisecond batch. We group trades that share the same exchange timestamp and aggressor side, while keeping price and quantity information. We deliberately avoid calling this batch a "parent market order," because the archive has no taker-order identifier to confirm that.

For a batch at time t_j, the reference book is the final top-of-book update strictly before the batch. Let m_j^- be its midpoint, τ the tick size, and $s_j = +1$ for an aggressive buy and $s_j = -1$ for an aggressive sell. The deepest signed reach is:

$$R_j = \max_{p \in batch_j} s_j \cdot (p - m_{j-})/\tau \tag{1}$$

At horizon h, the adverse maker-signed markout is

$$Y_j(h) = s_j \cdot [m(t_j + h) - m_{j-}]/\tau \tag{2}$$

A positive Y means the midpoint moved further in the aggressor's direction after the trade, which is bad news for a hypothetical passive maker sitting there. Using the strict-prior book keeps the trade itself from influencing the quote we measure reach against.

The strategy replay uses a coarser, but still causal, decision clock. A policy commits to a fixed quote for minute t, using the latest midpoint and state features available before that minute's flow starts. Within that minute, each aggressor-side/millisecond batch gets a deepest reach measured against the fixed quote reference. Any batch that reaches a quoted level counts as an execution opportunity. A common, deterministic binomial draw decides whether the policy's small resting order actually captures that opportunity, and we use the same draw across strategies whenever their exposure sets overlap.

3.3 Protocol Chronology

We developed the measurement hypothesis between 15–21 January. Before looking at any validation data, we fixed the depth-measurement protocol: the event unit, the alignment, the endpoint, the inference unit, the exclusions, and the pass rule. A successful validation run authorized exactly one final holdout. Before opening that holdout, we hashed the protocol, the estimator, the runner, the reporting code, the dependencies, and the

validation table. A later temporal and cross-instrument extension was frozen separately, before we downloaded or looked at any March outcomes.

Our first strategy protocol reserved 22 March–10 April. But Binance's official daily bookTicker archive ends on 30 March, so that window was infeasible. We kept a record of the missing dates and the failed checksum requests. After confirming what archive data was actually available, but before computing any replacement-period PnL or policy rankings, we fixed an archive-constrained design: 2–11 March for calibration and 12–30 March for evaluation. Because our measurement extension had already looked at a different endpoint on 2–21 March, we cannot call the strategy period an untouched market holdout. The honest protection here is narrower: the replacement dates, policies, endpoints, success rule, sensitivities, and executable hashes were all fixed before we opened any strategy outcomes.

4. Depth-Conditioned Adverse Selection

4.1 Cumulative Quote Exposure

For quote depth d, horizon h, and nonnegative weights w_j, we define

$$\eta(d,h) = \frac{\sum_j w_j Y_j(h) \cdot \mathbb{1}\{R_j \geq d\}}{\sum_j w_j \cdot \mathbb{1}\{R_j \geq d\}} \tag{3}$$

Equal batch weighting is primary and aggressive-quantity weighting is a declared sensitivity. The exposure set is cumulative. An event reaching 20 ticks can reach a sufficiently small resting quote at 0.5, 1.5, and every intermediate tested depth, subject to unobserved queue priority.

This differs from exclusive sweep-severity bins. If reach is partitioned into bins B_k with probability π_k and conditional markout μ_k, then a quote at threshold d_k has exposure

$$\eta(d_k) = \frac{\sum_{r \geq k} \pi_r \mu_r}{\sum_{r \geq k} \pi_r} \tag{4}$$

not merely μ_k. When deeper sweeps have larger markouts, using only the shallowest exclusive bin removes the most toxic events from the touch sample. In the development data, a superseded exclusive-bin analysis implied a next-to-touch ratio near 11; the corrected cumulative 60-second ratio is 3.608.

The frozen primary endpoint for day i is

$$D_i = \eta_i(1.5\ ticks, 10s) - \eta_i(0.5\ ticks, 10s) \tag{5}$$

A stage passes when three things hold: the two-sided day-level 95% Student-t interval sits above zero, at least 15 of 20 daily values are positive, and at least 18 days remain in the sample. For the mean $\overline{D}$ over n days, the day-level Student- t interval is the usual one:

$$\overline{D} \pm t_{n-1,0.975} \frac{s_D}{\sqrt{n}}, \qquad s_D^2 = \frac{1}{n-1} \sum_{i=1}^{n} (D_i - \overline{D})^2 \tag{6}$$

This treats each day as one exchangeable inference unit. That is a deliberately conservative choice, compared with treating individual batches or events as independent.

We also compute lag-3 Newey-West HAC intervals as a dependence diagnostic. For a day-indexed series D_i, with sample mean $\overline{D}$ and de-meaned values $\mu_i = (D_i - \overline{D})$, the lag-L Newey-West long-run variance estimator is:

$$\hat{\omega}^2 = \hat{\gamma}_0 + 2\sum_{\ell=1}^{L}\left(1 - \frac{\ell}{L+1}\right)\hat{\gamma}_\ell, \qquad \hat{\gamma}_\ell = \frac{1}{n}\sum_{i=\ell+1}^{n} u_i\, u_{i-\ell} \tag{7}$$

with L = 3 throughout. The matching HAC interval is $\bar{D} \pm z_{0.975}\frac{\sqrt{\hat{\omega}^2}}{n}$. Since $\hat{\omega}^2 > \hat{\gamma}_0$ whenever daily values are positively autocorrelated, this HAC interval is generally at least as wide as the plain Student- t interval, which matches the positive day-to-day persistence we see in Figure 1B. The fact that both interval families agree in sign throughout Tables 2, 3, and 5 is itself a good sign – it means our reported significance is not just an artifact of ignoring that persistence.

One-second and 60-second horizons, quantity weights, and book-age restrictions are all declared secondary analyses. The prospective extension adds predeclared daily medians and symmetric total 1% trimmed means.

4.2 Measurement Results

The original BTCUSDT stages all support a positive depth difference:

Table 2. *Cumulative reach-conditioned adverse-markout difference (1.5 ticks minus 0.5 ticks, 10-second horizon) across the three original chronological stages. Bold row denotes the pre-registered final holdout.*

Stage	Days	Mean difference (ticks)	SD	95% t interval	HAC interval	Positive days
Development	7	29.165	1.976	[27.337, 30.993]	[27.874, 30.456]	7/7
Validation	20	31.385	2.501	[30.214, 32.555]	[29.763, 33.007]	20/20
Final holdout	**20**	**37.764**	**5.709**	**[35.092, 40.436]**	**[35.165, 40.363]**	**20/20**

All three original BTCUSDT stages support a positive depth difference. The sign is positive on 47 of 47 original days. The magnitude, though, is not stable: it rises across the three stages, and its spread grows too. Thus, this is best described as a replicated direction, not a fixed structural coefficient.

Table 3. *Prospectively frozen 20-day temporal (BTCUSDT) and cross-instrument (ETHUSDT) extension of the depth-conditioned markout difference, reported for the mean, a symmetric 1% trimmed mean, and the median.*

Instrument	Endpoint	Mean difference	95% t interval	HAC interval	Positive days
BTCUSDT	**Mean (primary)**	**42.963**	**[39.665, 46.260]**	**[40.179, 45.746]**	**20/20**
BTCUSDT	Symmetric 1% trim	43.062	[39.784, 46.340]	[40.289, 45.834]	20/20
BTCUSDT	Median	46.838	[40.649, 53.026]	[40.977, 52.698]	19/20
ETHUSDT	**Mean (primary)**	**22.469**	**[20.569, 24.368]**	**[20.473, 24.464]**	**20/20**
ETHUSDT	Symmetric 1% trim	22.467	[20.556, 24.378]	[20.460, 24.475]	20/20
ETHUSDT	Median	23.825	[20.396, 27.254]	[20.554, 27.096]	19/20

Our separately frozen extension also passes. The mean and trimmed results are nearly identical to each other. The two zero-median days both land on the same calendar date, 8 March, and neither median series ever goes negative. The sign stays positive on 67 straight BTCUSDT days, and on all 20 ETHUSDT extension days. Cross-instrument magnitudes are not directly comparable, though, because tick values, price levels, volatility, and liquidity all differ between the two, which lines up with the invariance caution from Kyle and Obizhaeva (2016) discussed in Section 2.2.

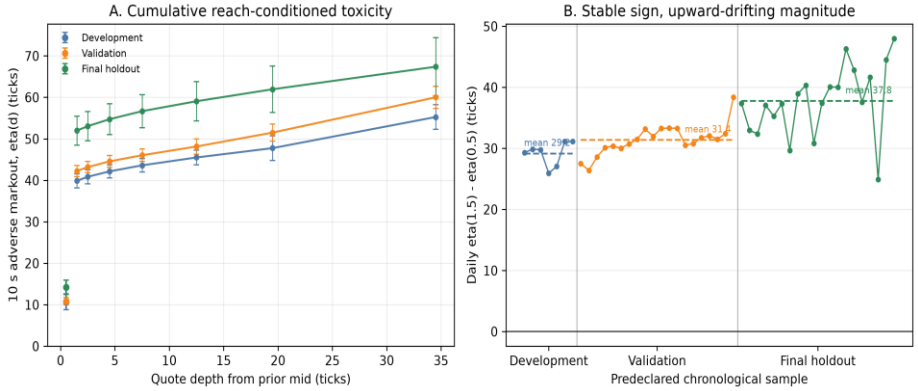


Figure 1. Day-level mean 10-second adverse-markout differences and 95% Student-t intervals across the original BTCUSDT stages. The touch point is separated from the curve beginning at 1.5 ticks because no intervening threshold was estimated. Daily differences reveal upward stage drift.

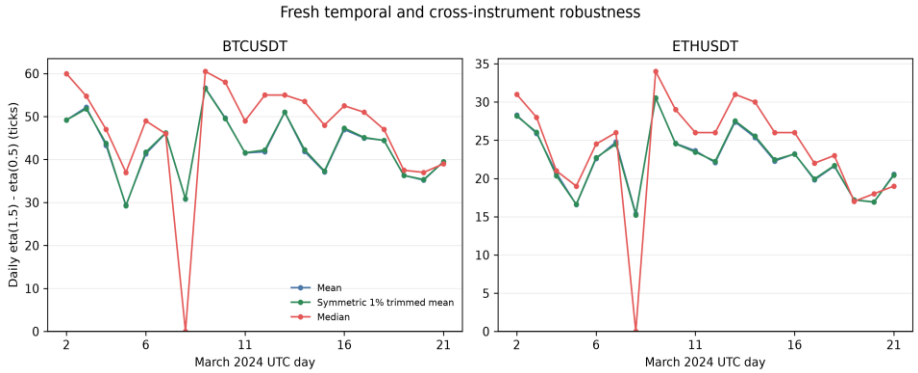


Figure 2. Prospectively frozen temporal and cross-instrument extension of the cumulative reach-conditioned adverse-markout difference. BTCUSDT and ETHUSDT are reported separately because their tick values, price levels, and liquidity differ.

The final-holdout secondary results keep the same direction. Mean differences come out to 30.946 ticks at a one-second horizon and 39.739 ticks at 60 seconds, both positive on all 20 days. All 14 of our declared 10-second weighting and book-age specifications are positive on every holdout day, even though the exact magnitudes shift around quite a bit.

4.3 What the Measurement Identifies

This result identifies a conditional exposure pattern, for a hypothetical, small, non-cancelling quote. It does not establish an actual fill probability, and it does not establish a causal, informed-trading mechanism either. Price continuation here could reflect real information, mechanical impact, liquidity being pulled endogenously, or just a shared reaction to news.

For control purposes, the important distinction is between the average impact of an aggressive trade, and the impact conditional on that trade actually reaching the selected quote. The second one is what matters for a passive fill. A schematic per-opportunity edge looks like this:

$$G(d,h) = reach_rate(d) \times queue_capture(d) \times [d \cdot \tau - \eta(d,h) \cdot \tau - fee] \quad (8)$$

This is not a complete control objective, but it does expose one important consistency condition: reach, queue capture, and adverse markout all need to refer to the same depth, the same event unit, and the same quote lifecycle. The no-quote action has value zero, and it needs to stay available whenever the estimated conditional edge turns negative.

5. Market-Making Models and Evaluation

5.1 AS and GLFT Benchmarks

Let S_t be the midpoint, q_t inventory in lots, σ_t fractional volatility, γ risk aversion, and $u = T - t$ remaining horizon. In price units, the implemented AS reservation price is

$$r_t = S_t - q_t(\gamma \cdot order_size)(S_t\sigma_t)^2 u \tag{9}$$

The symmetric half-spread before inventory skew is

$$h_t = 0.5(\gamma \cdot order_size)(S_t\sigma_t)^2 u + \frac{\ln\left[1+\frac{\gamma \cdot order_size}{\kappa/\tau}\right]}{\gamma \cdot order_size} \tag{10}$$

M0 uses calibration-mean volatility, and M1 uses causal, dynamic volatility instead. B3 is a stationary GLFT approximation, built with calibration-mean volatility, stationary side-specific arrival rates, and neutral liquidity. B1 just posts a fixed symmetric quote, and B2 adds a deterministic inventory skew on top.

For reference, the underlying Guéant-Lehalle-Fernandez-Tapia (2013) stationary approximation that B3 is drawn from gives us the asymptotic bid and ask depths, at neutral inventory, in closed form. Writing $\delta^{\pm,\infty}$ for the stationary half-spreads on each side, the closed-form solution is:

$$\delta^{\pm,\infty} = \frac{1}{\gamma \cdot \text{order_size}} \ln\left(1 + \frac{\gamma \cdot \text{order_size}}{\kappa/\tau}\right) + \frac{1}{2}\sqrt{\frac{\sigma^2\gamma \cdot \text{order_size}}{2\kappa/\tau A}\left(1 + \frac{\gamma \cdot \text{order_size}}{\kappa/\tau}\right)^{1+\frac{\kappa/\tau}{\gamma \cdot \text{order_size}}}} \tag{11}$$

where A is the stationary side arrival-intensity level, and κ is the exponential depth-decay parameter of the calibrated fill-rate curve, $\lambda(\delta) = Ae^{-\kappa\delta}$. Our implemented B3 actually uses the finite-horizon numerical solution rather than this asymptotic expansion, but the expansion is still useful; it clarifies why B3 counts as a stationary comparator. It holds A and κ fixed at their unconditional calibration values, rather than letting them respond to Hawkes intensity or liquidity regime, which is exactly the state-conditioning margin our HJB model is designed to add.

5.2 Hawkes Order Flow

Aggressive buy and sell batch times are fitted separately with exponential Hawkes intensities

$$\lambda_s(t) = \mu_s + \alpha_s \sum_{t_i^s<t} \exp[-\beta_s(t - t_i^s)] \tag{12}$$

Parameters are estimated on the ten-day calibration window. Evaluation events only update filtered intensity after they occur, so past evaluation flow is available to later decisions without any lookahead. The fitted branching ratios come out to 0.556 for buys and 0.586 for sells, both comfortably below one. For the single-exponential-kernel specification, the branching ratio has this closed form:

$$n_s = \alpha_s \int_0^\infty e^{-\beta_s v}\, dv = \frac{\alpha_s}{\beta_s} \tag{13}$$

Stationarity of the intensity process needs $n_s < 1$ on each side, and our fitted values satisfy this with some margin. The associated compensator, used for the held-out goodness-of-fit test below, is:

$$\Lambda_s(t) = \int_0^t \lambda_s(u)\, du \tag{14}$$

Stationarity alone does not mean the model is well specified. Held-out time-rescaling Kolmogorov-Smirnov (KS) statistics come out around 0.11, compared with critical values near 0.002, so every side-day rejects. This time-rescaling test applies a standard result: if λ_s is the true conditional intensity of the event times $t_1^s < t_2^s < \cdots$, then the rescaled inter-event increments

$$\tau_i = \Lambda_s(t_i^s) - \Lambda_s(t_{i-1}^s) \tag{15}$$

should be i.i.d. standard exponential, or equivalently, $1 - e^{-\tau_i}$ should be i.i.d. Uniform(0,1). The KS statistic detects departures from that uniformity. Rejection on every single side-day tells us something specific: our fitted exponential single-kernel Hawkes intensity is a useful causal forecast, but it is a misspecified one, not a validated model of how the process actually generates data. That is exactly why Section 5.3 treats it as a bucketed forecast input to the Bellman state, rather than as ground truth.

An exploratory 19-fold rolling comparison still finds something useful. Independent Hawkes improves held-out log likelihood over stationary Poisson by 0.325 per event, and a stable bivariate Hawkes adds another 0.0092 per event on top of the independent fits. Both gains are positive across all 19 folds. We treat Hawkes intensity as a causal clustering forecast here, not as a complete data-generating mechanism. This later bivariate diagnostic was never deployed in the frozen controller itself.

5.3 Liquidity Regimes and HJB Control

We divide calibration minutes into three states, using training-only liquidity quantiles. The fitted transition matrix is:

$$P = \begin{bmatrix} 0.509 & 0.318 & 0.174 \\ 0.317 & 0.371 & 0.312 \\ 0.175 & 0.311 & 0.514 \end{bmatrix} \tag{16}$$

Regime arrival multipliers are 0.635, 0.863, and 1.502. Estimated fill-decay rates are 0.00282, 0.00218, and 0.00218 per tick. The controller's bar-regression adverse-selection inputs are 0.493, 0.538, and 0.449 ticks per expected fill. These are scalar in quote depth. They are not the same estimand as the reach-conditioned 10-second markouts in Section 4, so their numerical levels are not compared as if they were interchangeable measurements.

The HJB value function is solved by backward induction over time n, inventory q, and liquidity state l:

$$V_n(q,l) = \max_{bid,ask} \{\mathbb{E}[spread\ edge - \eta_l \cdot fills - \gamma(q \cdot size \cdot S \cdot \sigma)^2] + \mathbb{E}[V_{n+1}(q + N_{bid} - N_{ask}, L_{n+1})]\} \tag{17}$$

Fill counts are conditionally Poisson, their difference is Skellam, both-sided fills earn spread revenue, and inventory transitions are truncated at a finite bound. If $N^{\text{bid}} \sim Poisson(\lambda^{\text{bid}})$ and $N^{\text{ask}} \sim Poisson(\lambda^{\text{ask}})$ are conditionally independent given the current state, the net fill imbalance $\Delta N = N^{\text{bid}} - N^{\text{ask}}$ has the Skellam probability mass function:

$$\Pr(\Delta N = k) = e^{-(\lambda^{\text{bid}}+\lambda^{\text{ask}})}\left(\frac{\lambda^{\text{bid}}}{\lambda^{\text{ask}}}\right)^{k/2} I_{|k|}\left(2\sqrt{\lambda^{\text{bid}}\lambda^{\text{ask}}}\right), \quad k \in \mathbb{Z} \tag{18}$$

where $I_{|k|}$ is the modified Bessel function of the first kind of order |k|. We impose the truncated inventory bound by reflecting or absorbing mass from $|q + \Delta N| > 20$ back onto the boundary state, before the backward-induction step runs. The terminal condition at n = N charges an explicit liquidation cost on any inventory left over:

$$V_N(q, l) = -|q| \cdot \text{size} \cdot \left(\frac{1}{2}\text{spread}_l + \text{impact}(|q|)\right) \tag{19}$$

Therefore, the controller pays both a half-spread crossing cost and a size-dependent impact term to liquidate at the horizon. It never gets to carry inventory through T for free.

Each side can choose no quote, or a book level from 0, 1, 2, 4, 8, 16, 32, 64, 96, 128, 192, 256, 384, corresponding to depths of 0.5 through 384.5 ticks, in a one-tick-spread minute.

The Bellman state is (n, q, l). Volatility and buy/sell Hawkes intensities enter as training-bucketed surface parameters, held fixed through each 30-minute solve, and refreshed each minute in a receding horizon. Thus, the HJB does condition on a clustering forecast, but it never evolves intensity inside the Bellman state itself, and it does not value the option to simply wait for that intensity to decay.

M2 keeps Hawkes intensity but masks liquidity variation. M3 keeps liquidity, but replaces Hawkes features with stationary arrivals. M4 uses both together. This gives us a genuine eight-policy ablation, rather than just comparing one proposed model against one weak baseline.

5.4 Execution and Inference

Every policy uses a 0.001 BTC order, a 30-minute planning horizon, a 20-lot inventory bound, zero maker fee, and 1% queue capture in the primary run. The quote stays fixed for one minute, and gets replenished immediately after a captured crossing. This is a non-cancelling, stale-quote stress test, not a low-latency market-making simulation, and this distinction is important for interpreting the results. Market-making PnL is:

$$MM\ PnL = Spread\ PnL + Adverse\text{-}Selection\ PnL - Fees - Liquidation\ Cost \tag{20}$$

Directional inventory PnL is reported separately. Our co-primary endpoints are MM PnL per hour, and MM PnL per USD 1 million of executed turnover. We average 20 common queue seeds within each day, and the 19 market days, not the seeds or the event opportunities, are our inference units. Paired M4−M0 and M4−B3 contrasts report Student-t, Bonferroni familywise, lag-3 Newey-West, Holm-adjusted, and sign evidence, all together.

Following the multiple-testing logic from Harvey and Liu (2015) and Bailey and López de Prado (2014), discussed in Section 2.5, the familywise interval for a single contrast, tested at level α across m pre-declared endpoints, uses the Bonferroni-adjusted critical value:

$$\bar{\Delta} \pm z_{1-\alpha/(2m)}\frac{s_\Delta}{\sqrt{n}} \tag{21}$$

The Holm step-down procedure orders the m endpoint p-values $p_{(1)} \leq \cdots \leq p_{(m)}$, and rejects hypothesis (k) only if $p_{(j)} \leq \alpha/(m - j + 1)$ for every $j \leq k$. Both corrections widen the nominal interval compared with the

unadjusted paired t-interval, and our frozen success rule in Section 5.4 requires the corrected interval, not the unadjusted one, to clear zero.

Our frozen success rule requires at least 17 usable days, at least one endpoint whose familywise interval clears zero against both M0 and B3, minimum activity, bounded inventory and drawdown ratios, and positive directions under four required capture and fee sensitivities. Sensitivities cannot rescue a failed primary contrast; that is a hard rule.

6. Strategy Results

6.1 Ranking and Paired Tests

All eight policies have negative mean market-making PnL:

Table 4. *Mean daily performance decomposition across all eight quoting policies under the frozen replay. Bold row denotes the proposed full HJB controller (M4).*

Policy	MM PnL / hour	MM PnL / USD 1m	Spread PnL / day	Adverse selection / day	Mean abs. inventory
B1 naive symmetric	−8.824	−652.24	1.96	−210.94	10.38
B2 inventory skew	−8.887	−649.02	1.66	−212.13	10.04
B3 stationary GLFT	−4.990	−683.62	66.28	−184.35	5.56
M0 classical AS	**−4.389**	**−664.44**	**70.05**	**−173.89**	**7.50**
M1 dynamic-volatility AS	−4.949	−686.19	63.62	−180.67	7.75
M2 Hawkes-only HJB	−5.062	−679.32	64.16	−183.93	6.09
M3 liquidity-only HJB	−5.015	−693.17	63.49	−182.14	6.22
M4 full HJB	−4.885	−672.02	65.78	−181.36	6.20

All eight policies end up with negative mean market-making PnL. The principal finding is that Inventory-aware policies quote wider than B1 and B2, capture more gross spread, and cut down absolute toxicity, but none of them actually overcomes adverse selection under this replay. Turnover normalization is not a universal ranking tool either; the naive policies lose more per hour, while somehow appearing less negative per USD 1 million, simply because they trade roughly twice as much and carry a lot more inventory.

The frozen paired comparisons are:

Table 5. *Paired M4 contrasts against the classical AS (M0) and stationary GLFT (B3) benchmarks, with multiple-testing-adjusted and dependence-robust intervals.*

Endpoint	Contrast	Mean daily Δ	95% interval	Familywise interval	HAC interval	Positive days
MM PnL/hour	M4 − M0	−0.496	[−0.682, −0.310]	[−0.742, −0.251]	[−0.727, −0.266]	0/19
MM PnL/hour	M4 − B3	0.105	[0.038, 0.172]	[0.017, 0.193]	[0.037, 0.173]	15/19
MM PnL/USD 1m	M4 − M0	−7.583	[−11.470, −3.697]	[−12.716, −2.451]	[−12.536, −2.631]	2/19
MM PnL/USD 1m	M4 − B3	11.596	[6.653, 16.539]	[5.069, 18.123]	[6.417, 16.775]	16/19

M4 beats stationary GLFT on both endpoints, but it is worse than classical AS on both endpoints too. Therefore, it fails our co-primary improvement gate. M4 turns over 1.10 times as much as M0, while carrying 0.83 times its mean absolute inventory. Notably, the negative AS comparison is not an inactivity problem, and it is not an inventory-bound violation either. The HJB actually controls inventory more tightly than M0, it just earns less net edge from all that extra execution.

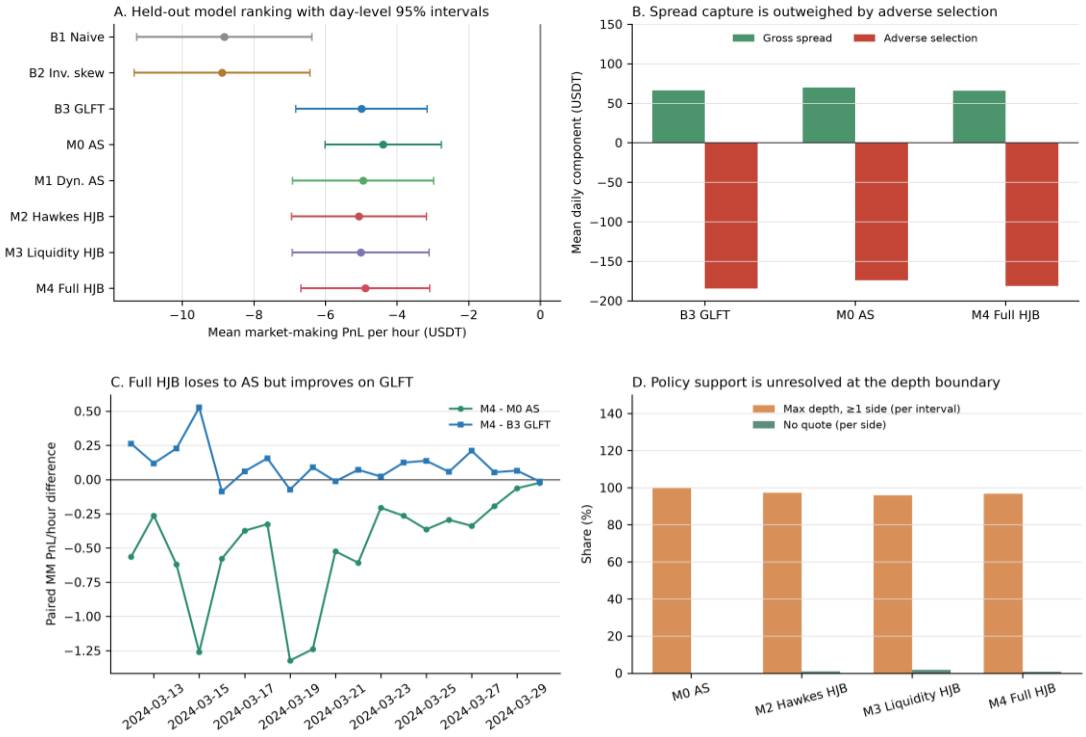


Figure 3. Day-level ranking, PnL decomposition, paired contrasts, and action-support diagnostics. Queue seeds are averaged before day-level intervals. In panel D, maximum depth is the share of intervals with at least one side at the threshold, while no quote is the share of individual sides. The panel is a model-validity diagnostic, not a performance result.

6.2 Sensitivity and Policy Support

All nine of our declared sensitivity configurations completed, using 20 common seeds each. These are descriptive, seed-level results over a continuous 19-day path; they're not a second, independent day-level inference exercise. M4−M0 MM PnL/hour stays negative in every single configuration, ranging from −0.911 at 2% capture, to −0.236 with a 40-lot inventory bound. M4−B3 is positive in seven of nine configurations, but it flips at 2% capture and at the 10-lot bound. No co-primary endpoint stays positive against both benchmarks, under all four sensitivities required by our frozen success rule.

The more fundamental limitation here is policy support. M0 reaches or exceeds the maximum replay threshold on every quote side. M4 chooses its maximum HJB action on at least one side in 96.7% of quote intervals, and picks no quote at all on only 0.8% of sides. Its median retained bid and ask depths both sit at 384.5 ticks. In effect, this experiment compares two bounded policy families against each other, it does not locate an unrestricted HJB optimum.

That boundary-seeking behavior comes from an unusual execution geometry. The empirical tail of event reach is almost flat across the tested depth support, which produces κ near 0.0022 per tick. More than four million

evaluation batches that reach the touch also reach the deepest level, because the quote stays fixed for a full minute while the market can travel a long way from its initial reference point. Event reach is a valid exposure opportunity under that lifecycle, but it is not the same thing as a historical queue-fill probability for a continuously updated maker.

7. Integrated Interpretation

7.1 What Hawkes Contributes

Our frozen comparison does not support the claim that Hawkes information is irrelevant. M2 and M4 consume separate buy and sell intensities, so directional imbalance can shift quote skew, even in cases where a common intensity multiplier barely moves a myopic depth optimum.

We ran a post-outcome structural diagnostic, reconstructing every executable, retained M4 action, and varying the frozen policy inputs across their empirical bucket range. Scaling buy and sell intensity together changed either HJB action in 20–25% of states. Holding total intensity fixed, but moving directional imbalance across its empirical range, changed either action in roughly 89% of states, and shifted the quote center materially. In the retained seed-42 traces we used for policy reconstruction, 76.45% of individual quote sides sit at the ceiling, and both sides sit at the ceiling in 56.04% of intervals. These post-outcome numbers help clarify the denominator behind the frozen 96.7% any-side interval rate, but they are not averaged over all 20 queue seeds, and this diagnostic has no confirmatory standing. What it does do is rule out one specific mechanical explanation for the null result: the side-specific Hawkes channel is not inert.

Thus, the evidence supports a more limited conclusion. Hawkes intensity forecasts clustering and directional pressure well enough, but the deployed HJB cannot turn that forecast into robust superiority, when the execution environment is poorly identified at the actions it actually selects. The held-out likelihood improvement makes Hawkes a genuinely useful feature. The failed residual tests just mean we should not treat its exact parametric intensity as structural truth. This reading lines up with Jusselin (2020): proving that a well-posed Hawkes-augmented control problem exists does not by itself guarantee that a coarsened, depth-discretized implementation of that control will recover a usable advantage over a stationary benchmark.

7.2 The Measurement-Control Mismatch

The depth study and the strategy study end up estimating two different things. The HJB uses a bar-regression impact coefficient, multiplied by mean trade size, producing one ηl per liquidity regime. The measurement study, on the other hand, conditions directly on whether an aggressive batch reaches the hypothetical quote, and estimates $\eta(d, h)$ right at that quote depth. This second approach is closer to the real cost conditional on a passive fill, though it still lacks queue rank and cancellation information.

It would be a mistake to subtract the replay's aggregate adverse-selection PnL per fill from ηl, and call the difference a calibration error. Replay PnL nets simultaneous bid and ask fills, uses a fixed minute-start quote, and marks at the minute close. The measurement endpoint, by contrast, uses side-specific event exposure and a 10-second horizon. These numbers simply are not commensurate with each other.

The structural mismatch is real, though. The controller's cost stays scalar in depth, while our separately validated exposure estimate changes sharply between 0.5 and 1.5 ticks. The HJB then goes on to select quotes around 340–384 ticks, far outside the roughly 34-tick region our depth surface actually covers. Neither study identifies toxicity at the actions the controller actually prefers. Thus, the combined evidence points toward building

$\eta(d, l, h)$ and queue-aware capture as future inputs; it does not show that bolting on an extrapolated depth penalty would have made M4 profitable or superior. This is the same alignment condition Alfonsi and Blanc (2016) raise abstractly for Hawkes-driven impact models in Section 2.3: closed-form or numerically stable control in this model family assumes the impact/markout kernel and the fill-intensity kernel are calibrated over the same depth and event support, and that is not the case here.

7.3 Standard for Benchmark Improvement

The lesson here is fundamentally methodological. A proposed market-making controller should not get declared better just because it beats a naive strategy, or a stationary model, inside a simulator built from its own assumptions. A credible benchmark should do seven things:

1. Estimate forecasts on past data, and update them causally;
2. Expose every policy to the same realized market opportunities;
3. Align execution probability and adverse selection by quote depth;
4. Use market days, not events or random seeds, for performance inference;
5. Include strong AS and GLFT comparators, plus component ablations;
6. Report inventory, turnover, drawdown, PnL decomposition, no-quote use, and action-boundary rates; and
7. Withhold optimality claims whenever the selected policy sits on the edge of its tested support.

Under this standard, M4 shows partial improvement over stationary GLFT, but not robust betterment of classical market making. This is a benchmark diagnosis, not a trading recommendation, and we note this distinction explicitly.

8. Limitations and Future Work

First, bookTicker only gives us Level-I data. Event reach cannot reveal queue rank, displayed depth behind the best quote, partial fills, hidden liquidity, or cancellation. Our strategy's queue-capture probability is a declared stress parameter; it is not an estimate of historical fill probability.

Second, the measurement assumes a small quote stays present right up until the aggressive batch arrives, and the strategy fixes quotes for a full minute with immediate replenishment. A live market maker could cancel or reprice, using imbalance, signed flow, latency, and queue state. Our replay is intentionally designed to expose stale-quote risk, but it is not a low-latency implementation.

Third, side/millisecond batches are observable, but they do not have to equal actual parent market orders. Millisecond timestamps do not order messages within a tie. We use the same unit consistently for event reach and for Hawkes filtering, but the underlying process interpretation stays bounded.

Fourth, our measurement covers two cryptocurrency futures on one venue, and the strategy covers one contract over 29 days. BTC effect sizes drift across stages, and ETH magnitudes look different too. Generalizing across assets, venues, fee tiers, and volatility regimes remains untested, and we do not test the invariance concerns from Kyle and Obizhaeva (2016) directly.

Fifth, our original strategy holdout turned out to be infeasible, because the official archive ended early. We froze the replacement strategy interval before computing any policy PnL, but its overlap with the already-studied measurement extension makes it an outcome-blind redesign, not a genuinely untouched market holdout.

Sixth, our exponential Hawkes specifications improve forecast likelihood, but they fail residual adequacy tests. M4 also buckets continuous volatility and side intensities, holding them fixed during each Bellman solve. A marked, order-book-dependent, or nonparametric flow state could be more faithful to reality, following, for example, Mucciante and Sancetta (2024), or the fully coupled Hawkes-HJB construction from Jusselin (2020), but any added complexity needs to earn its place out of sample.

Seventh, the action grid is binding, and it extends far beyond the depth where we actually measured adverse markouts. This paper cannot identify an unrestricted optimal quote, and it cannot claim that a depth-conditioned penalty would reverse the benchmark ranking.

Future work should be a separately frozen extension, not a post-outcome repair job. It should estimate side-specific $\eta(d,l,h)$ over a depth support that is actually relevant to the controller, test whether causal flow imbalance predicts both markout and short-horizon midpoint drift, and use Level-II or Level-III data to model queue position, cancellation, and partial execution. A corrected controller should then be compared against the present, frozen policies, on untouched dates and at least one additional instrument, with any resulting battery of trials evaluated using the multiple-testing corrections from Section 2.5, applied prospectively this time, not after the fact. The negative result reported here should be retained as the outcome of the initial evaluation.

9. Conclusion

This paper evaluates dynamic market making at two levels: measurement and control. Our cumulative, reach-conditioned evidence shows that aggressive events capable of reaching 1.5 ticks are followed by substantially larger adverse maker markouts, than the full exposure set reaching only 0.5 ticks. This direction passes a pre-specified validation and final holdout, and then replicates again in a prospective BTCUSDT and ETHUSDT extension. Its magnitude depends on the regime, and the estimand itself does not identify queue fills or informed trading directly.

Our strategy evaluation combines AS, GLFT, causal Hawkes forecasts, stochastic liquidity, and finite-inventory HJB control, all under common realized event reach. The full HJB improves on stationary GLFT, but underperforms classical AS, and every policy ends up with negative absolute market-making PnL. Directional Hawkes inputs do affect the policy, but the HJB stays boundary-seeking, in an environment where event reach is not the same as queue fill, and adverse-selection cost is not conditioned on quote depth.

Therefore, the evidence supports a restrained kind of betterment, an improvement to benchmark process, not a claim of strategy superiority. Dynamic state variables only turn out to be useful when the conditional execution and toxicity terms they control are measured on compatible exposure sets. A research-level market-making comparison needs to make that alignment visible, along with its causal timing, its action support, and its uncertainty, before treating additional model complexity as real progress.

Data and Reproducibility Statement

We fixed the measurement protocol and its prospective replication in writing, before looking at the corresponding data. We fixed the original strategy holdout, which later proved infeasible, and the archive-constrained replacement strategy design, in advance of the outcomes they govern.

Measurement outputs, manifests, and publication figures are retained in paper_outputs_depth/ and paper_outputs_extension/. Strategy seed-level, day-level, paired, sensitivity, point-process, and figure artifacts are retained in paper_outputs_strategy_v2/. The post-outcome Hawkes policy-channel diagnostic sits below paper_outputs_strategy_v2/post_outcome_hawkes_channel/, and it is not part of the frozen strategy outcome. Active code manifests match, and our automated test suite has 72 passing tests. The commands needed to rebuild every retained artifact from the processed inputs are documented alongside the codebase.

Declarations

Conflict of Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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